Data Collection

Keywords

Census: Measures every member of population
+ Accurate result — expensive / time consuming / destroy products.

Sampling units: Individuals of a population
Sampling frame: list of sampling units

Random Sampling

i) Simple Random Sampling

? same chance of being selected
random number / lottery sampling
+ bias free — need sampling frame

ii) Systematic Sampling

? take every k^{th} unit, $k = \frac{\text{Pop}}{\text{Sam}}$
Start at random number between 1 and k
+ quick to use — need Sampling Frame

iii) Stratified Sampling

? Sample represents group (strata) of a population
$\frac{\text{Sam}}{\text{Pop}}$ × Strata for each strata, choose randomly.
+ reflects population — Pop must be classified into strata.

Non Random Sampling

i) Quota Sampling

? like stratified, strata filled by interviewer
+ no Sampling Frame — non-random, potential bias

ii) Opportunity Sampling

? quota filled by those available at time
+ easy/cheap — Unlikely to be representative

Note, there are other advantages and disadvantages

Types of Data: qualitative non-numerical
quantitative numerical either discrete or continuous

Large Data Set

UK Stations — When?
Coastal — May – Oct '87 and '15
South — Only 6 months

International Stations
1) Perth, Australia
2) Beijing, China
3) Jacksonville, FL, USA — Oct '87, Oct '15

Data

Rainfall, "tr" means trace, treat as 0.025mm* in calculations
n/a means reading is not available, can't use in Sample.
Cloud cover: oktas, discrete value, integers 0-8
Max Gust: knots, 1 kn = 1.15 mph, Great Storm oct 15/16 '87

* updated from 0 since recording of video.

Location

KNOW YOUR CALCULATOR!

Mean

$\bar{x} = \frac{\Sigma x}{n}$ $\bar{x} = \frac{\Sigma fx}{\Sigma f}$

Listed Data

Position of $Q_1 = \frac{n}{4}$ (LQ) $Q_2 = \frac{n}{2}$ (Med) $Q_3 = \frac{3}{4}n$ (UQ)
If a decimal, round up ↑
If whole, find midpoint with next number

Grouped Data

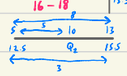
Position of $Q_1 = \frac{n}{4}$ $Q_2 = \frac{n}{2}$ $Q_3 = \frac{3n}{4}$
Percentiles eg. 57th percentile, $P_{57} = 0.57 \times N$
Deciles, 10% chunks, $D_3 = P_{30} = 0.3 \times N$
Do not round, use linear interpolation

Linear Interpolation

eg. weight, nearest kg

	true class limits	frequency	cumulative freq.
10 – 12	$9.5 \le w < 12.5$	5	5
13 – 15	$12.5 \le w < 15.5$	8	13
16 – 18	$15.5 \le w < 18.5$	7	20

$Q_2 = \frac{20}{2} = 10^{th}$



$Q_2 = 12.5 + \frac{5}{8} \times 3 = 14.375$ kg

Proportion

Spread

KNOW YOUR CALCULATOR!

Interquartile Range

F♥ $IQR = Q_3 - Q_1$ + Ignores extremes

Interpercentile Range

eg. 10th to 90th $IPR = P_{90} - P_{10}$

Variance, σ^2 / Standard Deviation, σ

$\sigma^2 = \frac{\Sigma x^2}{N} - \bar{x}^2$ MSMSM

$= \frac{\Sigma fx^2}{\Sigma f} - \bar{x}^2$ Note: $\Sigma fx^2 \neq (\Sigma fx)^2$

"mean of squares − square of mean"

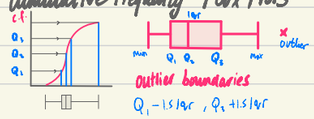
Coding

If $y = ax + b$

$\bar{y} = a\bar{x} + b$ $\sigma_y = a\sigma_x$

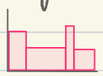
Representation

Cumulative Frequency / Box Plots



outlier boundaries
$Q_1 - 1.5 \times IQR$, $Q_3 + 1.5 \times IQR$

Histograms



continuous data
no gaps
frequency density = $\frac{\text{Freq}}{\text{Class width}}$
area = freq × k

Comparisons

1) Location 2) Spread context.

Regression + Correlation

KNOW YOUR CALCULATOR!

Product Moment Correlation Coefficient

PMCC, r $-1 \le r \le 1$
Measures strength + positivity/negativity of correlation

Regression Line

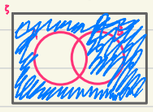
Line of best fit $y = a + bx$
a is y when x = 0
b is how much y changes with respect to x
✓ interpolation: estimate in data range
✗ extrapolation: estimate outside data range

Exponential / Non Linear Models

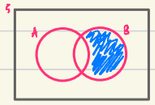
If $y = ab^x$ Take logs
$\ln y = \ln a + x \ln b$
If $y = ax^n$
$\ln y = \ln a + n \ln x$

Probability

Venn Diagrams

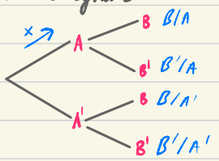


eg. $A' \cup B$ (not A)
Unions Shade all of both regions



eg. $A' \cap B$
Intersections Shade overlap

Tree Diagrams



Mutually Exclusive

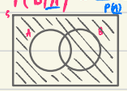
If A and B are mutually exclusive
$P(A \cap B) = 0$
$P(A \cup B) = P(A) + P(B)$

Independence

If A and B are independent
F $P(A \cap B) = P(A) \times P(B)$
F $P(A|B) = P(A)$ Can't tell from Venn

Conditional Probability

F♥ $P(B|A) = \frac{P(A \cap B)}{P(A)}$



Reduce the whole space to A

Addition Law

F $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

Discrete Uniform Distribution

Probabilities of outcomes all equal

eg. X is cloud cover, measured in oktas

x	0	1	2	3	4	5	6	7	8
P(X = x)	1/9	1/9	1/9	1/9	1/9	1/9	1/9	1/9	1/9

Binomial Distribution

$X \sim B(n, p)$
F $P(X = r) = \binom{n}{r} p^r (1-p)^{n-r}$

When to use?

Fixed number of trials, n
Fixed probability of success, p
Independent trials
Two outcomes, True/False

Cumulative Probabilities

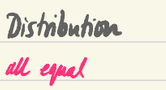
$P(X < 5) = P(X \le 4)$ (1,2,3,4 5) KNOW YOUR CALCULATOR!
$P(X > 3) = 1 - P(X \le 3)$ (4,5,6 ...)
$P(6 < X \le 10) = P(X \le 10) - P(X \le 6)$ (7,8,9,10)

Inverse Probabilities

Use tables!

Normal Distribution

Continuous Random Variable, Y
$Y \sim$
Point of inflection $\mu \pm \sigma$



} within } σ of μ

Finding Probabilities and Inverse Normal

eg. $P(Y > 122)$, $P(Y < a) = 0.2$ KNOW YOUR CALCULATOR!

Standard Normal Distribution

$Z \sim$
Coding, $Z =$
Missing μ, σ or both
Use coding, and sim eq. for both

Approximating Binomial as Normal

If n
If p
$\mu =$ $\sigma =$

Continuity Correction

If approximating binomial as normal
discrete →
eg. $P(X > 5) = P(Y > 5.5)$
$P(3 < X \le 11) = P(3.5 <$

Note: the 2 underlined values are incorrect in the video. These are the correct values.

Hypothesis Testing

Definitions

Null Hypothesis: H_0 = Assume true
Alternative Hypothesis: H_1 = true if H_0 is wrong
Significance Level: α, given threshold of likeliness.
One-tailed Test: when H_1: $p > k$ $p < k$
Two-tailed Test: H_1: $p \neq k$ have α value

Correlation Testing

* updated from r since recording the video. ρ is PMCC for population, r is PMCC for sample.

- H_0: ρ*
- H_1: ρ ρ ρ
- Compare your r with table
  - ↳ if your r is more extreme,
  - ↳ if your r is less extreme,

REMEMBER CONTEXT!

Binomial Testing

Test Statistic: number of successes observed.
- H_0: $p = k$ (given in Q)
- H_1: 1 tailed $p < k$ $p > k$ or 2 tailed $p \neq k$
- Assume H_0, i.e. $X \sim B(n, k)$
- Find $P(X \ge$ or $\le$ value in question) and going more extreme
  - ↳ if $P < \alpha$, reject H_0
  - ↳ if $P > \alpha$, Accept H_0, insufficient evidence.

REMEMBER CONTEXT!

Note, you should also be familiar with critical regions
Actual significance level is probability, P

Normal Testing – Sample Mean

Sample Mean, $\bar{x} \sim$
- H_0:
- H_1:
- Assume H_0, i.e.
- Find $P(\bar{x} >$ or $<$ mean of sample taken)
  - ↳ if $P < \alpha$,
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REMEMBER CONTEXT!

Data Collection

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Census:

\+ −

Sampling units:

Sampling frame:

Random Sampling

i) Simple Random Sampling

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\+ −

ii) Systematic Sampling

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\+ −

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Note, there are other advantages and disadvantages

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Large Data Set

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Only 6 months

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1)

2)

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Data

Rainfall, "tr" means trace, treat as 0.025mm* in calculations

n/a

Cloud cover

Max Gust

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Location

KNOW YOUR CALCULATOR!

Mean

$\bar{x}$ = $\bar{x}$ =

Listed Data

Position of Q_1 = (LQ) Q_2 = (Med.) Q_3 = (UQ)

If a decimal,

If whole,

Grouped Data

Position of Q_1 = Q_2 = Q_3 =

Percentiles eg. 57th percentile, P_{57} =

Deciles, 10% chunks, D_3 =

Do not round, use linear interpolation

Linear Interpolation

eg. weight, nearest kg	true class limits	frequency	cumulative freq.
10 – 12	≤ w <	5	
13 – 15	≤ w <	8	
16 – 18	≤ w <	7	

Q_2 =

$Q_2 = 12.5 +$

Spread

KNOW YOUR CALCULATOR!

Interquartile Range

F♥ IQR = +

Interpercentile Range

eg. 10th to 90th IPR =

Variance, σ^2 / Standard Deviation, σ

σ^2 = MSMSM

= Note: $\Sigma fx^2 \neq (\Sigma fx)^2$

Coding

If $y = ax + b$

$\bar{y}$ = σ_y =

Representation

Cumulative Frequency / Box Plots

c.f. x

outlier boundaries

Histograms

continuous data

no gaps

frequency density =

area =

Comparisons

1) 2)

Regression + Correlation

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PMCC, r ≤ r ≤

Measures

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Line of best fit

a is

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If $y = ax^n$

Probability

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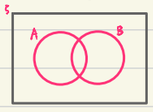
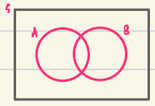


eg. $A' \cup B$

Unions



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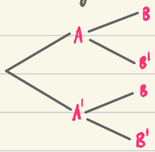

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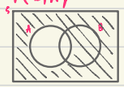

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x	
P(X = x)	

Binomial Distribution

$X \sim$

F $P(X = r)$ =

When to use?

F

F

I

T

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Continuous Random Variable, Y

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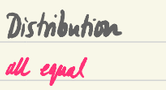

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If p

μ = σ =

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Test Statistic:

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- H_1:
- Assume H_0, i.e.
- Find $P(X \geq$ or $\leq$ value in question)

and going

- ↳ if $P < \alpha$,
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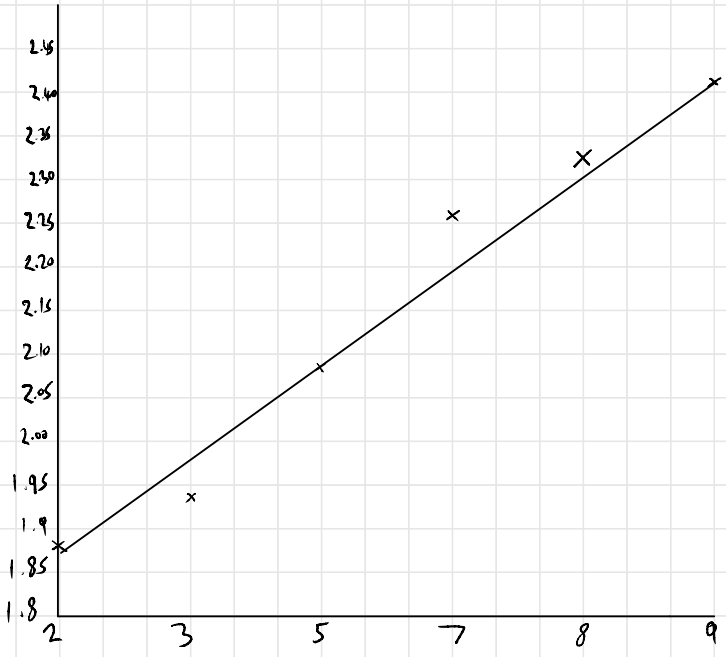
2.45
2.40
2.35
2.30
2.25
2.20
2.15
2.10
2.05
2.00
1.95
1.9
1.85
1.8
2
3
5
7
8
9

Statics. Normal distribution

1) female = $z = 1\frac{1}{3}$ $\sigma = 9$

male = $z = 1.2$ $\sigma = 10$